

Aryan Sahni

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SUMMARY

Research-driven Computer Science undergraduate at VIT Chennai with expertise in Artificial Intelligence, Machine Learning, Systems Programming, and Full-Stack Development. AI Intern with hands-on experience fine-tuning Veena TTS using the Indigo call data and designing a low-latency streaming architecture for a production-ready TTS service. Experienced in building full-stack AI platforms, Graph-RAG systems, and LLM-powered applications using TensorFlow, ONNX Runtime, LangChain, and modern AI frameworks. Published author at ImageCLEF Medical AI and multimodal image understanding, focused on developing scalable, data-driven solutions that bridge cutting-edge research with real-world applications.

EDUCATION

Vellore Institute of Technology, Chennai
B.Tech in Computer Science and Engineering

2023–2027
CGPA: 9.07/10

WORK EXPERIENCE

Data Science & AI Research Intern — AIONOS

[\[AIONOS\]](#)

- Engineered a low-latency streaming framework for a **3B-parameter LLM-based Text-to-Speech (TTS)** system, reducing **Time-to-First-Audio (TTFA)** by **92%** (19.43 s → 1.55 s) using **KV-cache incremental generation**, **chunk-wise SNAC decoding**, and streaming audio reconstruction.
- Designed and automated the end-to-end data preparation pipeline for **1,500+ minutes** of in-house airline call recordings, including annotation processing, audio preprocessing, **SNAC tokenization**, and training data generation for **QLoRA-based domain adaptation**.
- Developed a comprehensive benchmarking framework with **232 multilingual evaluation utterances**, automating latency, speech quality (**WER/CER**), speaker similarity, and GPU performance evaluation across baseline, fine-tuned, and streaming TTS models.

PROJECTS

EMAIL Bait Mail

[\[GitHub\]](#)

- Engineered a **Chrome extension** for real-time **phishing and scam detection** leveraging the **Gmail API**, **Flask microservices**, and DOM event tracking via **MutationObserver**.
- Fine-tuned an **AWS Bedrock Foundation Model** using a curated **phishing and scam email dataset** to improve threat classification reliability, enabling **LLM-powered email intent analysis**, phishing detection, and contextual response generation.
- Implemented asynchronous frontend-backend communication pipelines and dynamic content parsing for low-latency inbox monitoring and real-time email threat analysis.

Graph-RAG Knowledge Retrieval System

[\[GitHub\]](#)

- Built an end-to-end **Graph-RAG pipeline** using **LangChain**, **Ollama**, and **Zephyr LLMs**, achieving **78.8% response faithfulness** with **77% effective context utilization**.
- Enhanced retrieval quality through **knowledge graph traversal** and multi-hop reasoning, reducing **hallucinations to 12.6%** and improving factual grounding of generated responses.
- Engineered an automated benchmarking framework for **faithfulness**, **hallucination**, **graph coverage**, and **LLM-as-a-Judge evaluation**, enabling reproducible performance analysis across Graph-RAG configurations.

SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, SQL, Bash

Tools & Platforms: Git/GitHub, Linux, Docker, ONNX Runtime

Frameworks & Libraries: TensorFlow, PyTorch, LangChain, Hugging Face Transformers, Pandas, NumPy, Scikit-learn

AI/ML Expertise: Generative AI, Large Language Models (LLMs), Speech AI & TTS, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), Graph-RAG, Reinforcement Learning, Model Fine-Tuning, Model Evaluation

PUBLICATIONS

Sahni, Aryan, Rachit Gupta, et al. (2025). “Evaluating deep CNNs for multi-label concept detection in ROCov2 radiology image dataset by team LekshmiscopeVIT”. In: *CLEF2025 Working Notes, CEUR Workshop Proceedings, CEUR-WS.org, Madrid, Spain*. URL: https://ceur-ws.org/Vol-4038/paper_203.pdf.

Sahni, Aryan, Krishnesh Mathur, et al. (2026). “An Optimal Evaluation of Entanglement Depth in Variational Quantum Classifiers for Efficient Deployment of Machine Learning Applications”. In: *2026 International Conference on Electronic Systems and Intelligent Computing (ICESIC)*, pp. 320–325. DOI: [10.1109/ICESIC67389.2026.11496417](https://doi.org/10.1109/ICESIC67389.2026.11496417).